

GEFITINIB *intega* (Gefitinib 250mg Film-Coated Tablets)

PRESENTATION

Brown, round, biconvex, film-coated tablet impressed with “250” on one side and plain on the other. Each tablet contains 250 mg gefitinib.

INDICATIONS

GEFITINIB intega is indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with activating mutations of EGFR-TK.

DOSAGE AND ADMINISTRATION

The recommended dose of *GEFITINIB intega* is one 250 mg tablet once a day, taken with or without food. If a dose of *GEFITINIB intega* is missed, it should be taken as soon as the patient remembers. If it is less than 12 hours to the next dose, the patient should not take the missed dose. Patients should not take a double dose (two doses at the same time) to make up for a forgotten dose.

Where dosing of whole tablets is not possible, such as patients who are only able to swallow liquids, tablets may be administered as a dispersion in water. The tablet should be dropped into half a glass of drinking water (non-carbonated), without crushing, and the glass stirred until the tablet has dispersed (approximately 15 minutes) and the contents subsequently drunk immediately. The glass should be rinsed with a further half glass of water and the contents drunk. The liquid can also be administered via a nasogastric tube.

GEFITINIB intega is not recommended for use in children or adolescents as safety and effectiveness in these patient populations has not been studied.

No dosage adjustment is required on the basis of patient age, body weight, gender, ethnicity, or renal function or in patients with moderate to severe hepatic impairment due to liver metastases.

Dosage adjustment: Patients with poorly tolerated diarrhoea or skin adverse drug reactions may be successfully managed by providing a brief (up to 14 days) therapy interruption followed by reinstatement of the 250 mg dose.

CONTRAINDICATIONS

Known severe hypersensitivity to the active substance or to any of the excipients of this product.

WARNINGS AND PRECAUTIONS

When considering the use of *GEFITINIB integra* as first-line treatment for advanced or metastatic NSCLC, it is recommended that EGFR mutation assessment of the tumour tissue is attempted for all patients. When assessing the mutation status of a patient it is important that a well-validated and robust methodology is chosen to minimise the possibility of false negative or false positive determinations. Tumour samples which are used for the diagnosis of advanced NSCLC are the preferred sample type for EGFR mutation testing. A tumour sample should be collected and tested where possible. If a tumour sample is not available or evaluable, then circulating tumour DNA (ctDNA) obtained from a blood (plasma) sample may be used. Only robust, reliable, sensitive test(s) with demonstrated utility on ctDNA should be used for the determination of EGFR mutation status of ctDNA. It is not always possible to detect EGFR mutations using this sample type (0.2% false positives, 34.3% false negatives).

In the first line setting, *GEFITINIB integra* should not be used in preference to doublet chemotherapy in mutation-negative patients.

Interstitial Lung Disease (ILD), which may be acute in onset, has been observed in patients receiving gefitinib, and some cases have been fatal. If patients present with worsening of respiratory symptoms such as dyspnoea, cough and fever, *GEFITINIB integra* should be interrupted and prompt investigation initiated. If ILD is confirmed, *GEFITINIB integra* should be discontinued and the patient treated appropriately.

Liver function test abnormalities (including increases in alanine aminotransferase, aspartate aminotransferase, bilirubin) have been observed, uncommonly presenting as hepatitis. There have been isolated reports of hepatic failure which in some cases led to fatal outcomes. Therefore, periodic liver function testing is recommended. *GEFITINIB integra* should be used cautiously in the presence of mild to moderate changes in liver function. Discontinuation should be considered if changes are severe.

Cerebrovascular events have been reported in clinical studies of gefitinib. A relationship with gefitinib has not been established.

Substances that are inducers of CYP3A4 activity may increase metabolism and decrease gefitinib plasma concentrations. Therefore, co-medication with CYP3A4 inducers (e.g. phenytoin, carbamazepine, rifampicin, barbiturates or St John's Wort) may reduce efficacy.

International Normalised Ratio (INR) elevations and/or bleeding events have been reported in some patients taking warfarin. Patients taking warfarin should be monitored regularly for changes in Prothrombin Time (PT) or INR.

Drugs that cause significant sustained elevation in gastric pH may reduce plasma concentrations of gefitinib and therefore may reduce efficacy.

Patients should be advised to seek medical advice promptly in the event of developing:

- severe or persistent diarrhoea, nausea, vomiting or anorexia

These symptoms should be managed as clinically indicated.

Patients presenting with signs and symptoms suggestive of keratitis such as acute or worsening: eye inflammation, lacrimation, light sensitivity, blurred vision, eye pain and/or red eye should be referred promptly to an ophthalmology specialist.

If a diagnosis of ulcerative keratitis is confirmed, treatment with *GEFITINIB integra* should be interrupted, and if symptoms do not resolve, or recur on reintroduction of *GEFITINIB integra*, permanent discontinuation should be considered.

GEFITINIB integra should be used with caution in patients with a history of keratitis, ulcerative keratitis or severe dry eye. Recent corneal surgery and contact lens wearing are known to be independent risk factors for ocular toxicity including corneal erosion.

An increased risk of cerebral haemorrhage in adult patients with NSCLC receiving gefitinib has not been established.

Where gefitinib and vinorelbine have been used concomitantly, indicate that gefitinib may exacerbate the neutropenic effect of vinorelbine.

Gastrointestinal perforation has been reported in patients taking gefitinib. In most cases this is associated with other known risk factors, including increasing age, concomitant medications such as steroids or NSAIDs, underlying history of GI ulceration, smoking or bowel metastases at sites of perforation.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

In vitro studies have shown that the metabolism of gefitinib is predominantly via CYP3A4.

Co-administration with rifampicin (a known potent CYP3A4 inducer) have reduced mean gefitinib AUC by 83% of that without rifampicin.

Co-administration with itraconazole (a CYP3A4 inhibitor) resulted in an 80% increase in the mean AUC of gefitinib. This increase may be clinically relevant since adverse experiences are related to dose and exposure. Although interaction studies with other CYP3A4 inhibitors have not been performed it is expected that drugs such as ketoconazole, clotrimazole, ritonavir would also inhibit gefitinib metabolism.

Co-administration of ranitidine at a dose that caused sustained elevations in gastric pH (≥ 5), resulted in a reduced mean gefitinib AUC by 47%.

INR elevations and/or bleeding events have been reported in some patients taking warfarin.

PREGNANCY AND LACTATION

There are no data from the use of gefitinib in pregnant or breast-feeding women. Studies in animals have shown reproductive toxicity. Animal studies also indicate that gefitinib and certain metabolites pass into rat's breast-milk.

Women of childbearing potential must be advised to avoid becoming pregnant, and breast-feeding mothers must be recommended to discontinue nursing while receiving *GEFITINIB integra* therapy.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

During treatment with gefitinib, asthenia has been reported and those patients who experience this symptom should observe caution when driving or using machines.

POSSIBLE ADVERSE REACTIONS

The most commonly reported adverse drug reactions (ADRs), are diarrhoea and skin reactions (including rash, acne, dry skin and pruritus). ADRs usually occur within the first month of therapy and are generally reversible.

Table 1 Adverse drug reactions by system organ class and frequency

Metabolism and nutrition disorders	Very Common	Anorexia mild or moderate (CTC grade 1 or 2)
Eye disorders	Common	Conjunctivitis, blepharitis, and dry eye*, mainly mild (CTC grade 1)
	Uncommon	Corneal erosion, reversible and sometimes in association with aberrant eyelash growth Keratitis
Vascular disorders	Common	Haemorrhage, such as epistaxis and haematuria
Respiratory, thoracic and mediastinal disorders	Common	Interstitial lung disease often severe (CTC grade 3 or 4). Cases with fatal outcomes have been reported

Gastrointestinal disorders	Very Common	Diarrhoea, mainly mild or moderate (CTC grade 1 or 2) and less commonly, severe (CTC grade 3 or 4) Vomiting, mainly mild or moderate (CTC grade 1 or 2) Nausea, mainly mild (CTC grade 1) Stomatitis, predominantly mild (CTC grade 1)
	Common	Dehydration, secondary to diarrhoea, nausea, vomiting or anorexia
	Uncommon	Dry mouth*, predominantly mild (CTC grade 1) Pancreatitis Gastrointestinal perforation
	Very Common	Elevations in alanine aminotransferase, mainly mild to moderate
Hepatobiliary disorders	Common	Elevations in aspartate aminotransferase, mainly mild to moderate Elevations in total bilirubin, mainly mild to moderate
	Uncommon	Hepatitis
	Very Common	Skin reactions, mainly a mild or moderate (CTC grade 1 or 2) pustular rash, sometimes itchy with dry skin, including skin fissures, on an erythematous base
Skin and subcutaneous tissue disorders	Common	Nail disorder Alopecia
	Rare	Allergic reactions, including angioedema and urticaria Bullous conditions including Toxic epidermal necrolysis, Stevens Johnson syndrome and erythema multiforme Cutaneous vasculitis
	Common	Asymptomatic laboratory elevations in blood creatinine Proteinuria Cystitis
	Rare	Haemorrhagic cystitis
Renal and urinary disorders	Common	Asymptomatic laboratory elevations in blood creatinine Proteinuria Cystitis
General disorders and administration site conditions	Very Common	Asthenia, predominantly mild (CTC grade 1)

* This adverse reaction can occur in association with other dry conditions (mainly skin reactions) seen with gefitinib.

OVERDOSE

There is no specific treatment in the event of overdose of *GEFITINIB* *intega*. Adverse reactions associated with overdose should be treated symptomatically; in particular severe diarrhoea should be managed as clinically indicated

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Gefitinib is a selective inhibitor of the epidermal growth factor receptor (EGFR) tyrosine kinase, commonly expressed in solid human tumours of epithelial origin. Inhibition of EGFR tyrosine kinase activity inhibits tumour growth, metastasis and angiogenesis and increases tumour cell apoptosis.

Clinical characteristics of never smoker, adenocarcinoma histology and female gender have been shown to be independent predictors of positive EGFR mutation status. Asian patients also have a higher incidence of EGFR mutation-positive tumours.

Resistance

Most NSCLC tumours with sensitizing EGFR kinase mutations eventually develop resistance to gefitinib treatment with a median time to disease progression of 1 year. In about 60% of cases, resistance is associated with a secondary T790M mutation for which T790M targeted EGFR TKIs may be considered as a next line treatment option. Other potential mechanisms of resistance have been reported following treatment with EGFR signal blocking agents including bypass signalling such as HER2 and MET gene amplification and PIK3CA mutations. Phenotypic switch to small cell lung cancer has also been reported in 5-10% of cases.

Pharmacokinetic properties

Following intravenous administration, gefitinib is rapidly cleared, extensively distributed and has a mean elimination half-life of 48 hours. Following oral dosing in cancer patients, absorption is moderately slow and the mean terminal half-life is 41 hours. Administration of gefitinib once daily results in 2 to 8-fold accumulation with steady state exposures achieved after 7 to 10 doses. At steady state, circulating plasma concentrations are typically maintained within a 2 to 3-fold range over the 24-hour dosing interval.

Absorption:

Following oral administration of gefitinib, peak plasma concentrations of gefitinib typically occur at 3 to 7 hours after dosing. Mean absolute bioavailability is 59% in cancer patients. Exposure to gefitinib is not significantly altered by food.

Distribution:

Mean volume of distribution at steady state of gefitinib is 1400L indicating extensive distribution into tissue. Plasma protein binding is approximately 90%. Gefitinib binds to serum albumin and α 1-acid glycoprotein.

Metabolism:

In vitro data indicate that CYP3A4 is the major P450 isozyme involved in the oxidative metabolism of gefitinib.

In vitro studies have shown that gefitinib has limited potential to inhibit CYP2D6.

Gefitinib shows no enzyme induction effects in animal studies and no significant inhibition (*in vitro*) of any other cytochrome P450 enzyme.

Three sites of biotransformation have been identified in the metabolism of gefitinib: metabolism of the N-propylmorpholino-group, demethylation of the methoxysubstituent on the quinazoline and oxidative defluorination of the halogenated phenyl group. Five metabolites have been fully identified in faecal extracts and the major component was O-desmethyl gefitinib, although this only accounted for 14% of the dose.

In human plasma 8 metabolites were fully identified. The major metabolite identified was O-desmethyl gefitinib, which was 14-fold less potent than gefitinib at inhibiting EGFR stimulated cell growth and had no inhibitory effect on tumour cell growth in mice. It is therefore considered unlikely that it contributes to the clinical activity of gefitinib.

The production of O-desmethyl gefitinib has been shown, *in vitro*, to be via CYP2D6. In poor metabolisers, no measurable levels of O-desmethyl gefitinib were produced. The range of gefitinib exposures achieved in both the extensive and the poor metaboliser groups were wide and overlapping but the mean exposure to gefitinib was 2-fold higher in the poor metaboliser group. The higher average exposures that could be achieved by individuals with no active CYP2D6 may be clinically relevant since adverse experiences are related to dose and exposure.

Elimination:

Gefitinib total plasma clearance is approximately 500 mL/min. Excretion is predominantly via the faeces with renal elimination of drug and metabolites accounting for less than 4% of the administered dose.

Special populations:

In population based data analyses in cancer patients, no relationships were identified between predicted steady state trough concentration and patient age, body weight, gender, ethnicity or creatinine clearance.

PRE-CLINICAL SAFETY DATA

Not applicable.

PHARMACEUTICAL PARTICULARS

Special precautions for storage

Store below 30°C.

Shelf-life

4 years.

Pack size

30 tablets

Product Registration Holder in Malaysia

Intega Sdn Bhd

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Manufacturer

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